The Three Layers — What Each One Is For

A first look: an AI agent is built from three stacked layers, plus a data subsystem on the side.

L3 Skills — "What to do"

Holds the job descriptions for tasks: the goal, the input/output format, and the rules for when to act, when to stop, and when to loop.

Like an employee's playbook / SOP.

Add a Skill → the agent can do one more kind of task (logical scaling)

L2 Harness — "How the work gets done"

Turns a Skill's intent into concrete, runnable steps. Manages memory, context, tools, code generation, safety checks and audit logs.

Like a project manager + the runtime framework.

The single place where Skills and Scaffold meet (the decoupling point)

L1 Scaffold — "Where it runs, safely"

Provides the isolated place to execute: sandbox, shell, network, storage and model-serving capacity, with security boundaries.

Like a secure cloud computer.

Add an instance → more throughput (physical scaling)

Data Subsystem $\mathcal{D}$

What the agent knows about your data (kept off to the side, mostly offline)

- D_1 fetch — read the slice you need, on demand
- D_2 semantic summary — pre-digest sources offline
- D_3 governance — remember how to use the data
- D_4 lifetime — freshness, scope & cost budget
- Ω workspace — run history, artifacts, LLM wiki

Key idea: each kind of growth (more skills / more machines / more data) is added on a different axis, so they don't slow each other down.